

Real-Time Whole-Body Contact Estimation of Continuum Robots Using Multiplexed Fibers for Embodied Actuation and Sensing to Quantify Interactions

Zecai Lin^{1,2,3}, Jingyuan Xia^{1,2,3}, Zheng Xu^{1,2}, Yun Zou¹, Cheng Zhou^{1,2}, Jiafan Chen¹, Lucas Tat-Long Tong¹, Shaoping Huang¹, Huanghua Liu^{1,2}, Weidong Chen^{1,2,*}, Guang-Zhong Yang^{1,*}, Anzhu Gao^{1,2,*}

Abstract— Millimeter cable-driven continuum robots exhibit shape conforming, dexterous manipulation capabilities in constrained environments. They are increasingly used for narrow space and endoluminal intervention. For delicate manipulation, quantifying force interaction between the robot and its surrounding environment is important for both shape adjustment and avoiding damages to luminal structures. In this work, we propose a real-time, whole-body contact estimation framework for small-scale continuum robots, based on actuation fibers and model-informed neural networks. The physical relationship among external body contact, internal actuation and shape sensing of the continuum robot is formulated based on rod theory, and the body contact estimation is treated as an inverse problem given the actuation tension profile and robot shape as inputs. The contact position and force are estimated using a neural network, and a generative adversarial network (GAN)-based data augmentation strategy is proposed to reduce the need for large amounts of real data from the continuum robot under external forces. Additionally, an automatic data acquisition platform is developed to efficiently collect the small amount of required data. Experiments with notched continuum robots were conducted to demonstrate the general applicability and accuracy of the proposed approach. The results show that the mean estimation errors for the 3D contact position and contact force magnitude are 1.7 mm (2.3%) and 8.7 mN (5.8%), respectively, with an estimation frequency of 25 Hz. It paves the way for embodied integration using multiplexed fibers for the simultaneous actuation and sensing of millimeter-scale continuum robots, enabling safer operation in confined spaces through machine intelligence.

I. INTRODUCTION

Tendon-driven continuum robots are increasingly deployed in constrained environments owing to their high flexibility, dexterity, and natural shape conformability [1]–[3]. Designed with large length-to-diameter ratios, these high-degree-of-freedom structures can naturally navigate tortuous lumina to reach deep-seated surgical targets [4], [5]. During endoluminal procedures, however, inadvertent wall contact along the flexible body is practically unavoidable. Such interaction may induce robot looping or even cause tissue perforation without accurate force feedback [6], [7].

Consequently, real-time in-situ whole-body contact quantification is essential for safety-critical shape adaptation.

Current contact estimation methods for continuum robots fall into three main categories. Model-based force sensing formulates external loads using tension or kinematic states [8]–[14]; however, most rely on idealized frictionless assumptions and struggle under real friction conditions. Electronic methods, including resistive, capacitive, and piezoelectric sensors [15]–[18], provide localized feedback but remain too rigid or bulky to coat sub-millimeter bodies smoothly. Optical sensing using Fiber Bragg Gratings (FBGs) offers high biocompatibility and compact integration [19]–[23]. Nevertheless, existing FBG techniques primarily target tip force estimation, contact localization without force magnitudes [24], or passive instrument tracking lacking active cable actuation [25].

To bridge these gaps, we propose a real-time, whole-body contact estimation framework driven by embodied machine intelligence. Single-core optical fibers (SCFs) embedding FBGs are multiplexed to perform simultaneous mechanical driving and tension sensing. Integrated with a Cosserat rod physical model and a Generative Adversarial Network (GAN) data augmentation scheme, our model-informed neural network achieves accurate 3D contact location and force recovery at 25 Hz across dual-robot grasping, RCM-constrained force avoidance, and airway phantom interventions.

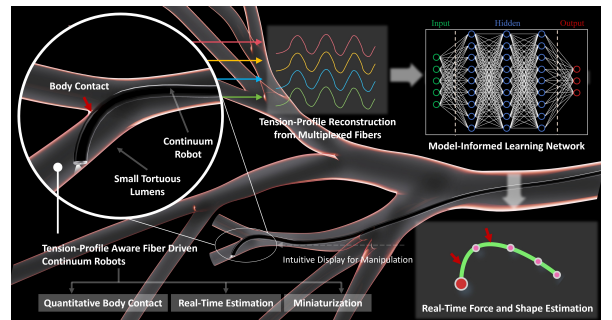


Fig. 1. Overview of the proposed real-time whole-body contact estimation method. A flexible continuum robot moving in a tortuous and narrow lumen may inevitably contact with the surrounding environment, and the contact force needs to be fed back to ensure safe operation. A model-informed learning-based method was used to predict the whole-body contact using multiplexed fibers for simultaneous actuation and sensing, which have the advantages of quantifying body contact, enabling real-time estimation, and embodied integration for miniaturization.

¹The Institute of Medical Robotics, Shanghai Jiao Tong University, Shanghai, China.

²Department of Automation, Shanghai Jiao Tong University, and the Key Laboratory of System Control and Information Processing, Ministry of Education, Shanghai, China.

³These authors contributed equally: Zecai Lin, Jingyuan Xia.

II. METHODS

A. Embodied Sensor & Robot Configuration

To validate the framework across different kinematic freedoms, two cable-driven continuum robots were developed using Nitinol tubes with laser-cut symmetric notches:

Planar (1-DOF) Prototype: The outer tube features an outer diameter (OD) of 3.0 mm and inner diameter (ID) of 1.3 mm. Two single-core optical fibers (SCFs, OD 200 μm) with embedded FBG arrays are routed symmetrically through internal positioning blocks. These two fibers function dual-purpose as opposing driving cables and dynamic tension sensors for 2D contact estimation.

Spatial (2-DOF) Prototype: The outer tube possesses an OD of 3.5 mm and ID of 1.4 mm. A central 4-core Multicore Fiber (MCF, OD 400 μm) is aligned along the central axis to reconstruct continuous 3D shape profiles. Four orthogonal SCFs (OD 200 μm) with FBG arrays act simultaneously as quadruplex actuation cables and tension-profile sensors.

By replacing conventional metal driving cables with FBG-embedded optical fibers, active actuation and high-density state sensing are multiplexed into a sub-millimeter footprint without occupying the central working channel.

B. Real-time Whole Body Contact Estimation

The state of a continuum robot—influenced by actuation fibers and external forces—can be represented by its curvatures, which allow for full shape reconstruction. Based on this, we propose a model-informed neural network framework (Fig. 2a-b) that formulates contact estimation by mapping actuation FBG wavelength shifts, robot shape, curvatures, and bending angles to the contact position and

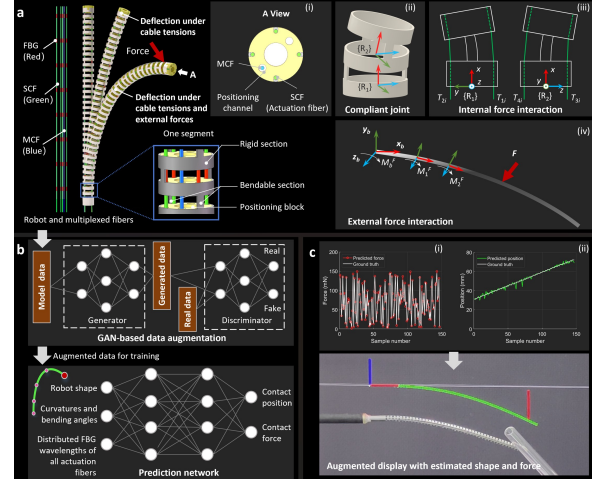


Fig. 2. Model-informed neural network for body contact estimation.

external force. Given the low-dimensional nature of these inputs and outputs, a Multilayer Perceptron (MLP) with non-linear activation functions and backpropagation training effectively handles the complex, non-linear mapping.

For 2D contact estimation, the MLP takes a 220-dimensional input—comprising FBG wavelength changes, 2D shape coordinates, and curvature distributions—and predicts a 3-dimensional output (1D arc-length contact position from the base and 2D contact force components within the bending plane).

Extending contact estimation from 2D to 3D significantly increases force distribution complexity. Collecting vast real-world training datasets is difficult and risks degrading the lifespan of sensors and robot structures. To address data scarcity, a Generative Adversarial Network (GAN, Fig. 2b)

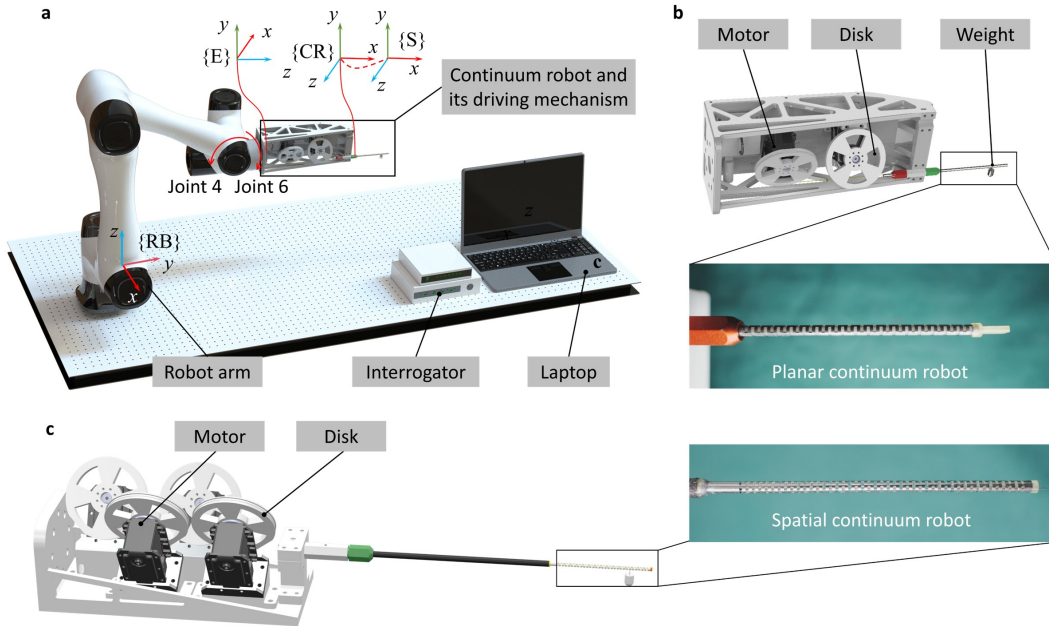


Fig. 3. The experimental setup for training data acquisition and contact estimation validation. (a) The continuum robot and its driving mechanism were fixed to the end of the 6-DOF robot arm. (b) Planar continuum robot and its driving mechanism. (c) Spatial continuum robot and its driving mechanism.

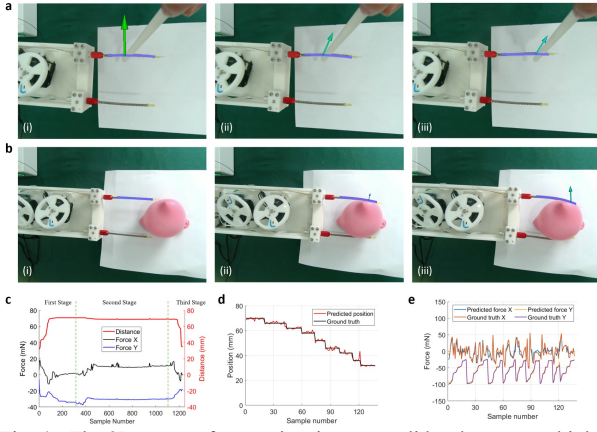


Fig. 4. The 2D contact force estimation was validated across multiple scenarios: (a) real-time AR visualization of external forces applied along the robot body, (b-c) grasping force estimation during a dual-robot balloon grasping task, and (d-e) quantitative accuracy evaluation against ground truth data for predicted contact position and force magnitude.

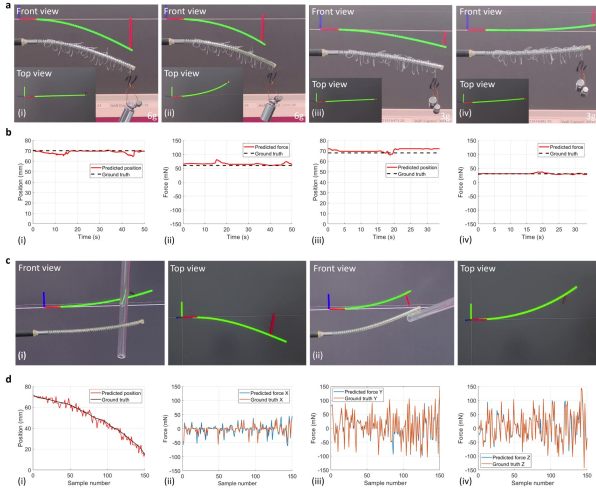


Fig. 5. Demonstrates 3D contact force estimation decoupled from active bending under 3 g/6 g weights (a-b), validates real-time estimation across diverse initial shapes (c), and evaluates quantitative accuracy against ground truth (d).

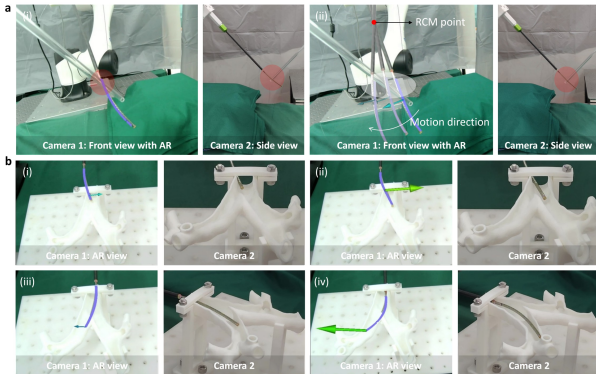


Fig. 6. Showcases potential clinical utility via RCM-constrained force-avoidance motion under high contact forces (a) and teleoperated endoluminal intervention inside an airway phantom (b).

augments the training dataset by aligning physics-based simulation data with real experimental data. The GAN generator takes model simulation data as input, while the

discriminator evaluates generated outputs against real data. Because manufacturing errors, assembly inconsistencies, and material variations cause domain shifts between physical models and real-world conditions, the GAN effectively learns these domain discrepancies, aligning simulation and real-data distributions to robustly inform and train the 3D contact estimation network. The model data was first used to validate the effectiveness of the prediction network, and the simulation results of 3D contact force estimation are shown in Fig. 2c. The position is the distance from the base of the robot to the contact point along the arc length of the robot. The estimated mean errors of the contact force magnitude and contact position were 5.5 mN, 3.7% and 0.9 mm, 1.2%, respectively.

III. EXPERIMENT AND RESULTS

A. Experiment Setup and Training Data Acquisition

To validate the method, contact force estimation was conducted on planar (1-DOF) and spatial (2-DOF) laser-cut NiTi continuum robots. Multi-core and single-core fibers (MCF/SCFs, FBGS International) were embedded to sense deformation. An FBG interrogator monitored wavelength shifts at 25 Hz. The robots were mounted on a 6-DOF arm (Dobot CR5) for data acquisition, with computing performed on an NVIDIA RTX A4000 GPU. Weights of different masses were individually attached to various positions along the continuum robot using a string to apply external forces. Since the gravitational force acts vertically downward, the ground-truth external force can be calculated in real time based on the pose of the 6-DOF robot arm. In Fig. 3a, $\{RB\}$, $\{E\}$, $\{CR\}$, and $\{S\}$ represent the reference frames associated with the robot arm, the end-effector of the robot arm, the continuum robot, and the MCF, respectively. The z -direction of $\{RB\}$ is parallel to that of gravity, and the external force produced by a weight with respect to $\{RB\}$ is $\mathbf{F}_{RB} = [0 \ 0 \ G]^T$, where G is the gravity of the weight. $\mathbf{F}_{RB}$ can be transformed from $\{RB\}$ to $\{S\}$ as $\mathbf{F}_s = (\mathbf{R}_{RB}^E \mathbf{R}_E^{CR} \mathbf{R}_{CR}^S)^{-1} \mathbf{F}_{RB}$, where $\mathbf{R}_{RB}^E$ is the rotation matrix of the pose of the 6-DOF robot arm; $\mathbf{R}_E^{CR}$ is the rotation matrix for the transformation matrix from $\{E\}$ to $\{CR\}$, which can be obtained in advance based on the assembly configuration; $\mathbf{R}_{CR}^S$ is the rotation matrix for the transformation matrix from $\{CR\}$ to $\{s\}$, and it is an identity matrix. Finally, the external force with respect to $\{S\}$ can be obtained.

B. 2D contact force estimation

For evaluation, 12,000 datasets were collected using the planar continuum robot (split into 80% training and 20% validation), and hyperparameter tuning selected a 4-hidden-layer MLP (600-300-150-75 neurons) to predict 2D contact force components and the arc-length position from a 220-neuron input. The model was validated across three key experimental scenarios: real-time in-plane contact sensing visualized via AR-assisted force arrows, dual-robot grasping through narrow gaps to estimate grasping force and potential

object stiffness, and static load testing with various attached weights across different joint angles (Figs. 4d–e). Overall, the proposed method achieved high accuracy, with mean errors of 7.1 mN (4.7%) for contact force magnitude and 0.8 mm (1.1%) for contact position.

C. 3D contact force estimation

To evaluate the spatial continuum robot, a 4-hidden-layer MLP (700-350-175-87 neurons) was trained on a combination of 36,000 real datasets and 100,000 GAN-augmented samples (generated from 1,000,000 physics-based simulation sets using a 124-400-124 generator and 124-60-1 discriminator), mapping a 430-neuron input (FBG wavelength shifts, reconstructed shape, curvatures, and bending angles) to 4 outputs representing the arc-length contact position and 3D force components (F_x , F_y , F_z). Experimental validation demonstrated the model's performance across four scenarios: active bending decoupling trials with 3 g/6 g weights where estimated force vectors remained gravity-aligned during 3D cable-driven motion, multi-state force sensing under varying initial shapes and contact positions, static weight load testing across different robot arm angles yielding low mean errors of 1.7 mm (2.3%) for position and 8.7 mN (5.8%) for force magnitude, and potential medical application phantoms featuring RCM-constrained force-avoidance motion to prevent excessive contact damage as well as teleoperated endoluminal intervention within a human airway phantom.

IV. CONCLUSION

This work employs FBG-embedded optical fibers as actuation cables, achieving dual continuous bending actuation and real-time tension sensing without occupying extra space. To address the challenge of data scarcity under complex 3D contact force states, a GAN framework combines simulation and real-world data to enrich the training set, substantially reducing real-robot data collection efforts. Extensive validation experiments demonstrate the effectiveness of the proposed network in predicting 3D external forces at arbitrary locations on the continuum robot, estimating real-time grasping forces using dual parallel robots, executing force-avoidance motions under RCM constraints, and sensing body contacts during endoluminal interventions.

REFERENCE

- [1] E. Amanov, T. D. Nguyen, and J. Burgner-Kahrs, "Tendon-driven continuum robots with extensible sections—A model-based evaluation of path-following motions," *Int. J. Robot. Res.*, vol. 40, no. 1, pp. 7–23, 2021.
- [2] C. Zhou, Z. Xu, Z. Lin, et al., "Submillimeter fiber robots capable of decoupled macro-micro motion for endoluminal manipulation," *Sci. Adv.*, vol. 10, no. 47, p. eadr6428, 2024.
- [3] T. Zhang, L. Yang, X. Yang, et al., "Millimeter-scale soft continuum robots for large-angle and high-precision manipulation by hybrid actuation," *Adv. Intell. Syst.*, vol. 3, no. 2, p. 2000189, 2021.
- [4] A. Gao, N. Liu, M. Shen, et al., "Laser-profiled continuum robot with integrated tension sensing for simultaneous shape and tip force estimation," *Soft Robot.*, vol. 7, no. 4, pp. 421–443, 2020.
- [5] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, "Continuum robots for medical applications: A survey," *IEEE Trans. Robot.*, vol. 31, no. 6, pp. 1261–1280, 2015.
- [6] P. E. Dupont, N. Simaan, H. Choset, et al., "Continuum robots for medical interventions," *Proc. IEEE*, vol. 110, no. 7, pp. 847–870, 2022.
- [7] Z. Lin, H. Wu, H. Jia, et al., "Fixed and sliding FBG sensors-based triaxial tip force sensing for cable-driven continuum robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2022, pp. 9593–9599.
- [8] J. M. Ferguson, D. C. Rucker, and R. J. Webster, "Unified shape and external load state estimation for continuum robots," *IEEE Trans. Robot.*, vol. 40, pp. 1813–1827, 2024.
- [9] K. P. Ashwin, S. K. Mahapatra, and A. Ghosal, "Profile and contact force estimation of cable-driven continuum robots in presence of obstacles," *Mech. Mach. Theory*, vol. 164, p. 104404, 2021.
- [10] J. Back, L. Lindenroth, K. Rhode, et al., "Three dimensional force estimation for steerable catheters through bi-point tracking," *Sens. Actuators A Phys.*, vol. 279, pp. 404–415, 2018.
- [11] K. Xu and N. Simaan, "An investigation of the intrinsic force sensing capabilities of continuum robots," *IEEE Trans. Robot.*, vol. 24, no. 3, pp. 576–587, 2008.
- [12] J. Back, L. Lindenroth, R. Karim, et al., "New kinematic multi-section model for catheter contact force estimation and steering," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2016, pp. 2122–2127.
- [13] P. Wang, Z. Xie, W. Xin, et al., "Sensing expectation enables simultaneous proprioception and contact detection in an intelligent soft continuum robot," *Nat. Commun.*, vol. 15, no. 1, p. 9978, 2024.
- [14] V. Aloï, K. T. Dang, E. J. Barth, et al., "Estimating forces along continuum robots," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 8877–8884, 2022.
- [15] J. B. Gafford, S. B. Kesner, R. J. Wood, and C. J. Walsh, "Force-sensing surgical grasper enabled by pop-up book MEMS," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2013, pp. 2552–2558.
- [16] U. Kim, D. H. Lee, W. J. Yoon, et al., "Force sensor integrated surgical forceps for minimally invasive robotic surgery," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1214–1224, 2015.
- [17] F. Ju, Y. Wang, Z. Zhang, et al., "A miniature piezoelectric spiral tactile sensor for tissue hardness palpation with catheter robot in minimally invasive surgery," *Smart Mater. Struct.*, vol. 28, no. 2, p. 025033, 2019.
- [18] B. Gil, B. Li, A. Gao, et al., "Miniaturized piezo force sensor for a medical catheter and implantable device," *ACS Appl. Electron. Mater.*, vol. 2, no. 8, pp. 2669–2677, 2020.
- [19] J. K. Sahota, N. Gupta, and D. Dhawan, "Fiber Bragg grating sensors for monitoring of physical parameters: A comprehensive review," *Opt. Eng.*, vol. 59, no. 6, p. 060901, 2020.
- [20] Z. Wu, A. Gao, N. Liu, et al., "FBG-based triaxial force sensor integrated with an eccentrically configured imaging probe for endoluminal optical biopsy," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 1625–1631.
- [21] T. Zhang, Z. Ping, and S. Zuo, "Miniature continuum manipulator with three degrees-of-freedom force sensing for retinal microsurgery," *J. Mech. Robot.*, vol. 13, no. 4, p. 041002, 2021.
- [22] N. J. Deaton, T. A. Brumfiel, A. Sarma, et al., "Simultaneous shape and tip force sensing for the coast guidewire robot," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3725–3731, 2023.
- [23] A. Gao, N. Liu, H. Zhang, et al., "Spiral FBG sensors-based contact detection for confocal laser endomicroscopy," *Biosens. Bioelectron.*, vol. 170, p. 112653, 2020.
- [24] X. T. Ha, D. Wu, C. F. Lai, et al., "Contact localization of continuum and flexible robot using data-driven approach," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6910–6917, 2022.
- [25] Q. Qiao, G. Borghesan, J. De Schutter, et al., "Force from shape—estimating the location and magnitude of the external force on flexible instruments," *IEEE Trans. Robot.*, vol. 37, no. 5, pp. 1826–1833, 2021.